1. 1 gram is equal to

(A) 10 milligrams
(B) 100 milligrams
(C) 1 000 milligrams
(D) 10 000 milligrams

2. 3.1415926 expressed to TWO significant figures is

(A) 3.1
(B) 3.14
(C) 3.2
(D) 31

3. 2 μm means

(A) 2×10^{6} m
(B) 2×10^{3} m
(C) 2×10^{-3} m
(D) 2×10^{-6} m

4. Which of the following has NO units?

(A) Velocity
(B) Relative density
(C) Energy
(D) Momentum

5. Which of the following instruments is BEST suited for measuring the diameter of a strand of hair?

(A) Ruler
(B) Vernier callipers
(C) Engineer's callipers
(D) Micrometer screw gauge

6. Which of the following is NOT an SI base unit?

(A) m
(B) kg
(C) A
(D) cm^{3}

7. Which of the following energy sources is/are renewable?

I. Solar radiation
II. Wind energy
III. Fossil fuels

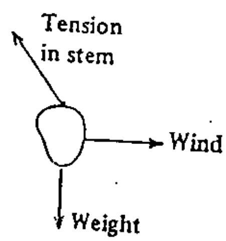
(A) I only
(B) II and III only
(C) I and III only
(D) I and II only

8. If an object is moving along a surface with a constant acceleration, the net force acting on the object is

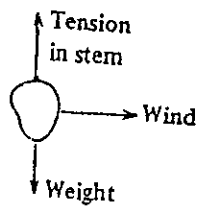
(A) zero
(B) constant
(C) increasing
(D) decreasing

9. A mango hangs from a long stem in a strong horizontal breeze. Which of the diagrams below BEST represents the forces acting on the mango?

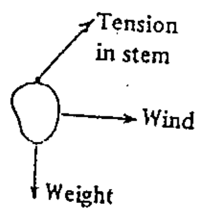
(A)



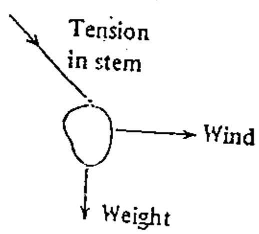
(B)



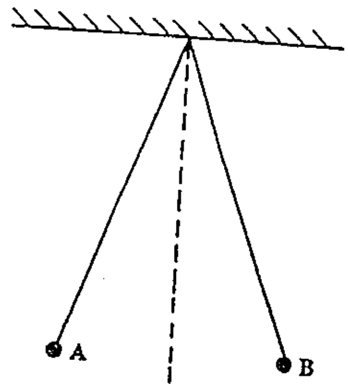
(C)



(D)



10.



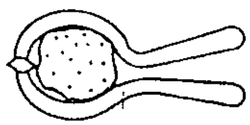
Two light, aluminium spheres, A and B, are suspended by insulating threads. If they come to rest as shown in the diagram, the force keeping them apart is

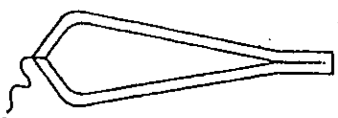
(A) gravitational force
(B) electrostatic force
(C) magnetic force
(D) centripetal force

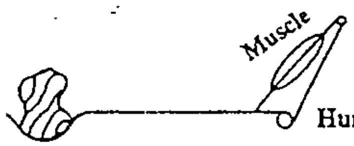
11. A stable well-designed racing car must have a

(A) low centre of gravity
(B) narrow wheel base
(C) sun roof
(D) long front

Item 12 refers to the following diagrams which depict some simple machines.

I 

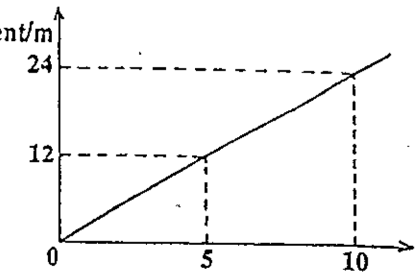
II 

III 

IV 

12. In which of these machines would the effort applied be less than the load?

(A) I and II only
(B) I and IV only
(C) II and IV only
(D) II and III only

13. 

The graph above shows how the displacement of a runner from a starting line varies with time. This runner is

(A) going at a steady speed
(B) going faster and faster
(C) not moving
(D) going slower and slower

14. A falling raindrop reaches a constant speed when

(A) there is no net force acting on it
(B) the pull of the earth on the raindrop is equal to the weight of the raindrop
(C) the upthrust due to the air is at a minimum
(D) the air surrounding the raindrop becomes saturated with water vapour

15. The constant unbalanced force acting on a body is at right angles to its velocity. Which of the following statements about this motion is correct?

(A) The body has no acceleration.
(B) The kinetic energy of the body is increasing.
(C) The body is slowing down.
(D) The body is moving in a circular path.

16. The kinetic energy of a body of mass, m, and velocity, v, is

(A) $\frac{m}{v}$

(B) mv

(C) $\frac{mv^2}{2}$

(D) mv^2

17. Which of the following gives the efficiency of a machine?

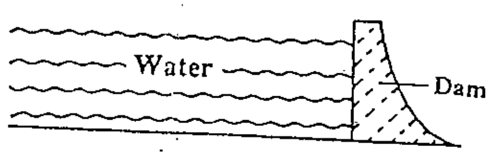
(A) $\frac{\text{Load x Distance moved by Effort}}{\text{Effort x Distance moved by Load}} \times \frac{100\%}{1}$

(B) $\frac{\text{Load x Distance moved by Load}}{\text{Effort x Distance moved by Effort}} \times \frac{100\%}{1}$

(C) $\frac{\text{Effort x Distance moved by Effort}}{\text{Load x Distance moved by Load}} \times \frac{100\%}{1}$

(D) $\frac{\text{Effort x Distance moved by Load}}{\text{Load x Distance moved by Effort}} \times \frac{100\%}{1}$

18.



The diagram above shows a dam. The pressure on the dam at the bottom of the reservoir depends on the

(A) depth of the water
(B) volume of water held by the dam
(C) mass of water held by the dam
(D) length of the reservoir

19. When liquids X and Y are taken from bottles at room temperatures and dabbed on the back of your hand, X feels cooler than Y. The MOST likely reason is that

(A) X has a higher specific heat capacity than Y
(B) X is a poorer conductor of heat than Y
(C) X evaporates more quickly than Y
(D) X is less dense than Y

20. Which of the following would explain why an ordinary 0 – 110 °C laboratory thermometer is not used to measure human body temperature?

I. The reading would change when the thermometer is taken from the patient's mouth.
II. It is not sensitive enough to measure small changes in temperature.
III. It does not have a large enough range.

(A) I only
(B) I and II only
(C) II and III only
(D) I, II and III

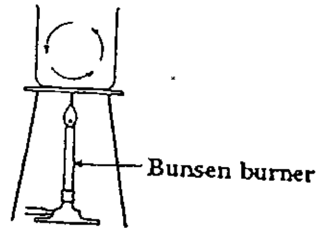
21. A fixed mass of gas at constant temperature is reduced in volume. Which of the following statements is NOT true about the gas?

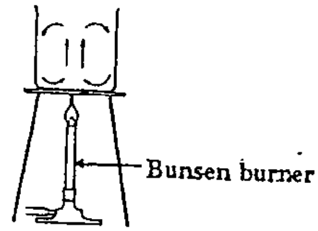
(A) The pressure of the gas increases.
(B) The average distance between the molecules decreases.
(C) The average speed of the molecules decreases.
(D) The molecules collide with the walls of the container more often.

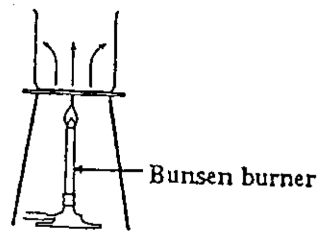
22. Which of the following statements about heat radiation is NOT true?

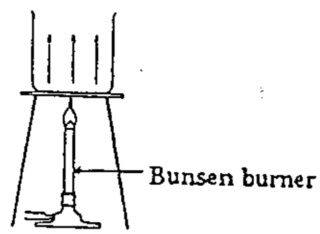
(A) Radiation is the transfer of heat by electromagnetic waves.
(B) A shiny surface is a better emitter of radiation than a dull black surface.
(C) Radiation can occur through a vacuum.
(D) Solar heat panels on houses are painted black to absorb more thermal radiation.

23. Which of the following diagrams BEST illustrates convection current in a liquid?

(A)


(B)


(C)


(D)


24. An experiment is set up with a smoke cell for demonstrating Brownian motion. The moving specks observed are

(A) smoke particles seen by reflected light
(B) smoke particles in constant vibration
(C) molecules of vibrating air
(D) molecules of carbon in random motion

25. Young's double slit experiment demonstrates that light

(A) has a particle nature
(B) produces sharp images
(C) travels in a straight line
(D) is a wave motion

26. Which of the following is the correct relation between the wavelength, λ, speed, υ, and frequency, f, of a wave?

(A) $\lambda = f\upsilon$

(B) $f = \frac{\lambda}{\upsilon}$

(C) $f = \frac{\upsilon}{\lambda}$

(D) $\lambda = \frac{f}{\upsilon}$

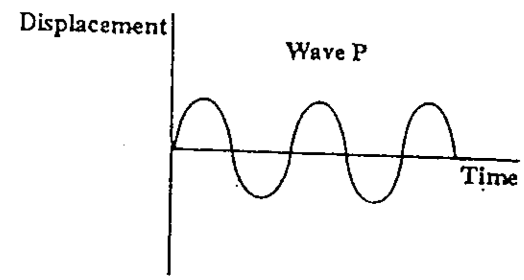
27. The table below lists the refractive indices for light of four different materials

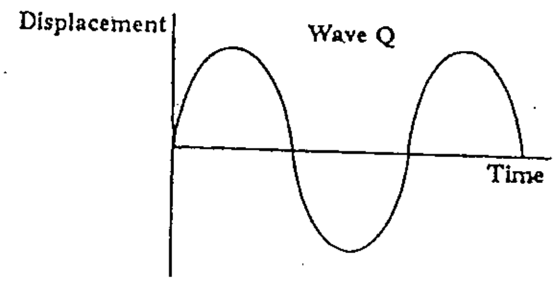
Material	Refractive Index
Air	1.0
Ice	1.3
Perspex	1.5
Diamond	2.4

In which medium would the light waves have the slowest speed?

(A) Air
(B) Ice
(C) Perspex
(D) Diamond

Item 28 refers to the following graphs (with axes having the same scales) of two sound waves, P and Q.





28. Which of the following statements is true?

(A) P is louder than Q but Q has a higher pitch.
(B) P is louder than Q and also has a higher pitch than Q.
(C) Q is louder than P but P has a higher pitch.
(D) Q is louder than P and also has a higher pitch than P.

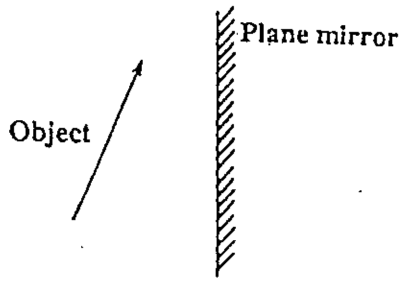
29. The normal hearing range of a young person is about

(A) 20 Hz to 2 000 Hz
(B) 20 Hz to 20 000 Hz
(C) 200 Hz to 2 000 Hz
(D) 200 Hz to 20 000 Hz

30. In the electromagnetic spectrum, X-rays are between

(A) gamma rays and ultra-violet rays
(B) ultra-violet rays and visible light
(C) visible light and infra-red rays
(D) infra-red rays and radio waves

31.



The diagram above represents an object placed in front of a plane mirror. Which of the following BEST represents the image produced by the plane mirror?

(A)

(B)

(C)

(D)

32. Which of the following does NOT sugg that light travels in straight lines?

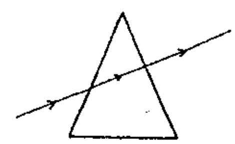
(A) Sunbeams streaming through tr
(B) A rainbow formation in the sky
(C) The formation of shadows
(D) Light from a projector on its way a screen

33. A ray of light in air strikes a glass block an angle of incidence of 0°. The light w be

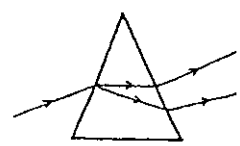
(A) undeviated
(B) totally reflected
(C) refracted at 90° to normal
(D) refracted at an unknown angle

34. Which of the diagrams BEST represer the passage of a beam of white light throu a triangular glass prism?

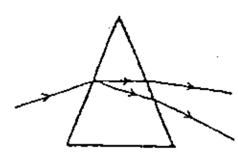
(A)



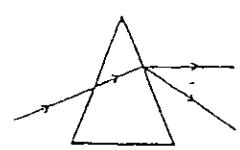
(B)



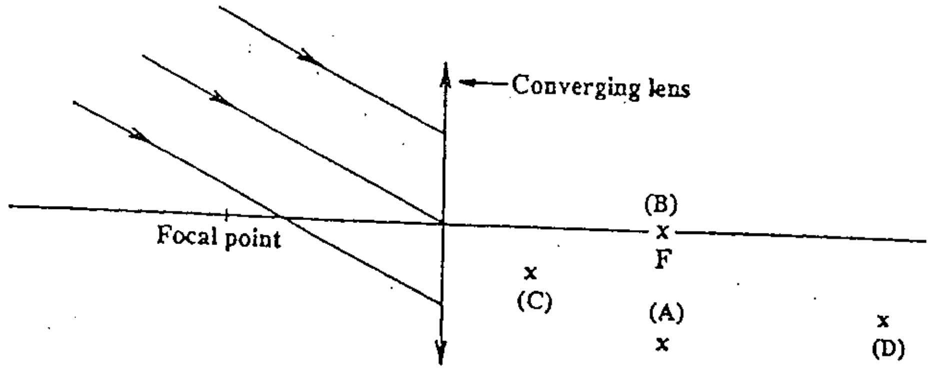
(C)



(D)

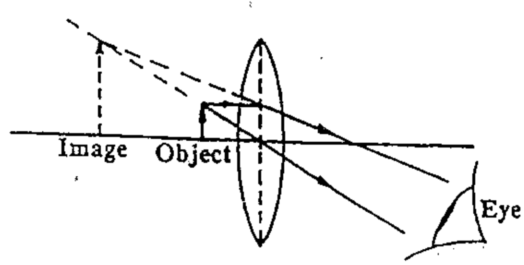


35.



In the diagram above, the incident rays of light from an object are parallel. Where will the image of the object be formed?

36.



The diagram above shows the formation of an image by a

(A) lens camera
(B) pin hole camera
(C) telescope
(D) magnifying glass

---

37. The current in a wire is one ampere if a charge of

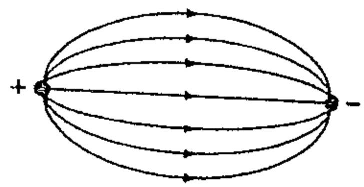
(A) 10 coulombs flows through it in 10 seconds
(B) 1 coulomb flows through it in 10 seconds
(C) 10 coulombs flow through it in 1 second
(D) 100 coulombs flow through it in 10 seconds

38. Which of the following relationships between electrical quantities is correct?

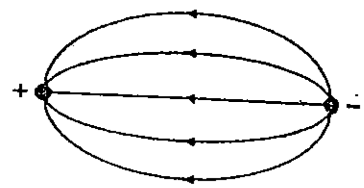
(A) $V = PI$
(B) $R = VI$
(C) $Q = \frac{E}{V}$
(D) $E = VI$

39. Which of the following diagrams represents the electric field existing between two oppositely charged point charges?

(A)

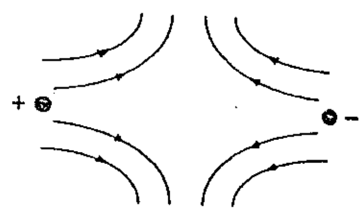


(B)



(C)

(D)



40. Which of the following ratios could represent the ampere?

I. $\frac{\text{Coulomb}}{\text{Second}}$

II. $\frac{\text{Volt}}{\text{Ohm}}$

III. $\frac{\text{Watt}}{\text{Volt}}$

(A) I only
(B) I and III only
(C) II and III only
(D) I, II and III

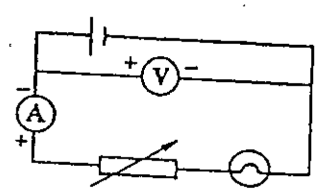
41. Which of the following units can be used measure quantities of energy?

I. Joule
II. Coulomb
III. Kilowatt-hour

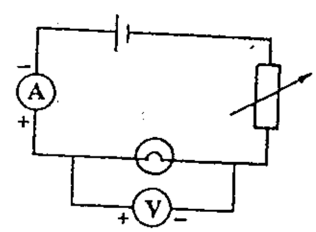
(A) I only
(B) I and II only
(C) I and III only
(D) II and III only

42. In which of the circuits below would the meters give readings which would enable you to determine the resistance of the filament lamp?

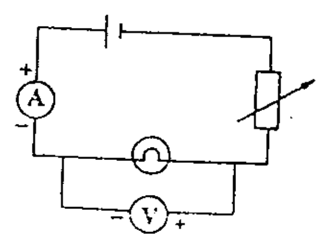
(A)



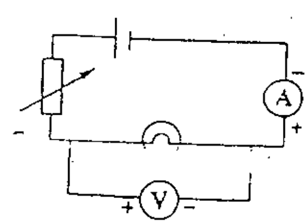
(B)



(C)

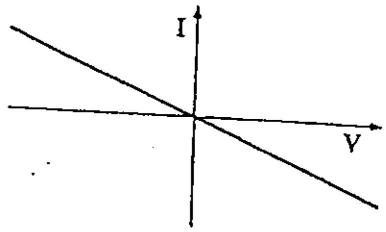


(D)

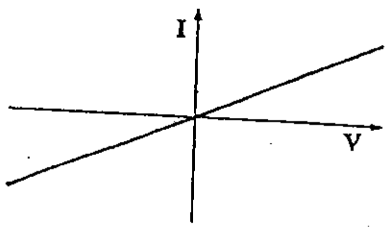


43. Which of the following is a representation of the current/p.d. relationship for a metallic conductor at a constant temperature?

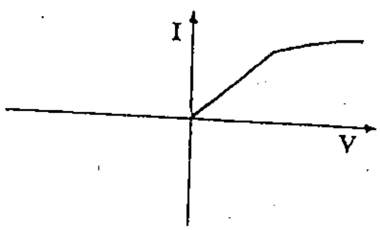
(A)



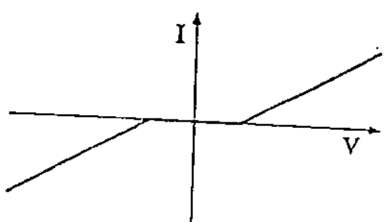
(B)



(C)



(D)

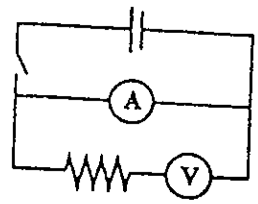


44. An appliance is connected to the mains by a cable which has three wires. What colour should the insulation be on the wire connected to the fuse?

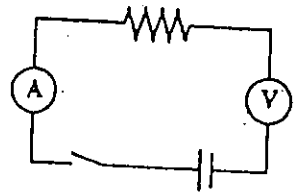
(A) Brown
(B) Black
(C) Blue
(D) Green/yellow

45. A student requires a circuit to measure the resistance of a resistor. Which of the circuits below is correctly connected?

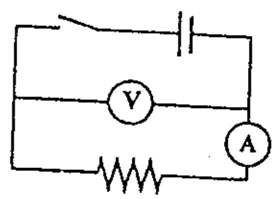
(A)



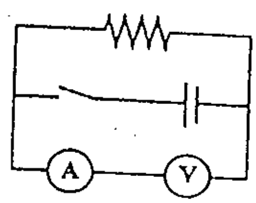
(B)



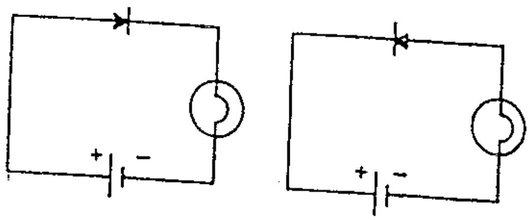
(C)



(D)



46.



An experiment was conducted using the circuit diagrams shown above. The same components were used and the bulb was lit to normal brightness in each case.

Which of the following statements would be correct?

I. The bulb is defective.
II. The battery is defective.
III. The diode is defective.

(A) I only
(B) III only
(C) I and II only
(D) II and III only

47. A semi-conductor diode produces half-wave rectification of alternating current. Which of the following statements would be true?

I. The current obtained has a constant value.
II. The current obtained flows only in one direction.
III. There are periods when no current flows from the source.

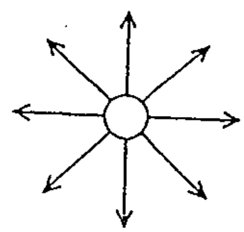
(B) II only
(C) I and III only
(D) II and III only

48. A suspended bar magnet always settl

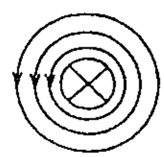
(A) with its N-pole pointing north
(B) with its N-pole pointing south
(C) perpendicular to the Ear magnetic field
(D) opposite to the direction from w it started

49. Which of the following shows the shap the magnetic field around a straight c ductor perpendicular to the page and ca ing current INTO the page?

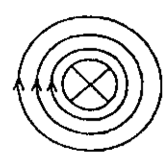
(A)



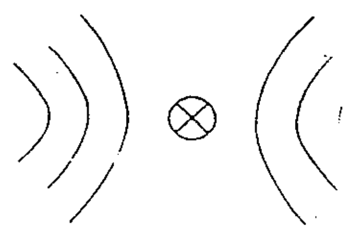
(B)



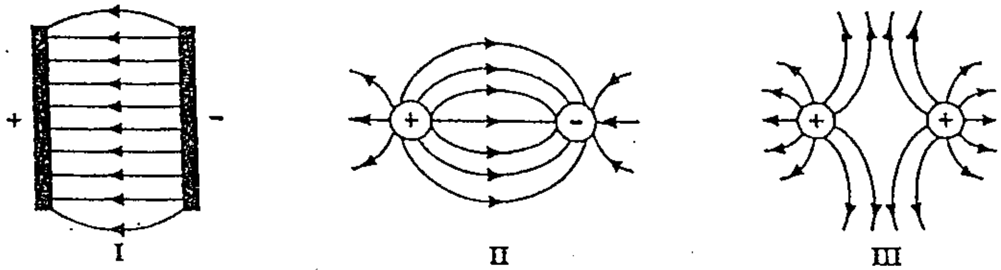
(C)



(D)



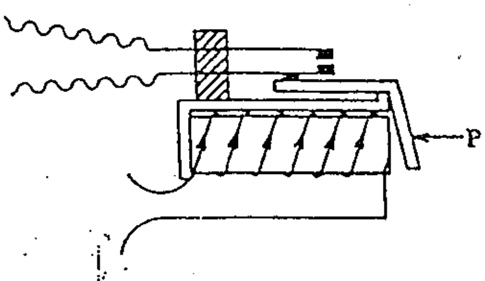
50.



The diagrams above represent the electric fields around and between point charges and parallel plates as drawn by a student. Which of the diagrams are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

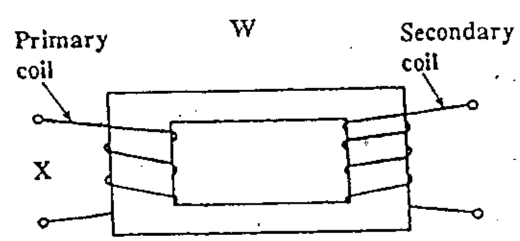
51.



The diagram above shows a typical relay. The part labelled P must be made of

(A) iron
(B) copper
(C) plastic
(D) brass

Item 52 refers to the following diagram.



Appropriate labels for W and X would be

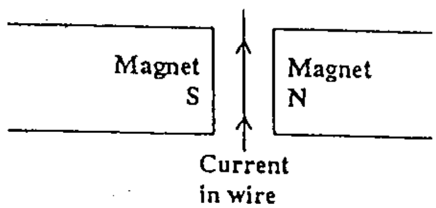
	W	X
(A)	step-down transformer	a.c. input
(B)	step-down transformer	d.c. input
(C)	step-up transformer	a.c. input
(D)	step-up transformer	d.c. input

GO ON TO THE NEXT PAGE

53. An electricity company supplies a.c. but not d.c. to the consumer. This is because

(A) its generators cannot produce d.c.
(B) most electrical appliances cannot run on d.c.
(C) a.c. can be generated at a higher voltage than d.c.
(D) a.c. can be transmitted more efficiently than d.c.

54.



The wire in the diagram above will move

(A) towards N
(B) towards S
(C) into the paper
(D) out of the paper

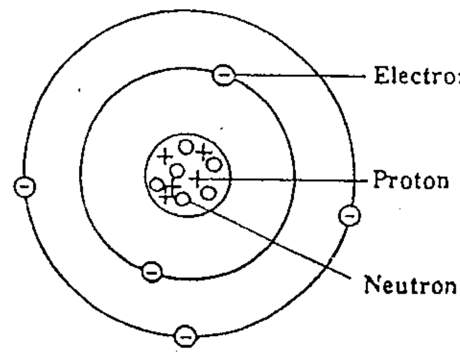
55. In the scattering experiment conducted by Geiger and Marsden, some of the alpha particles were deflected. The explanation for this phenomenon is that

(A) electrons have a small mass
(B) electrons have a small charge
(C) the metal foil was only a few atoms thick
(D) the nuclear charge and mass are concentrated in a small volume

56. The nuclide of an element, X, can be represented as $^{A}_{Z}X$. What does the letter A represent?

(A) Proton number
(B) Element symbol
(C) Neutron number
(D) Nucleon number

57. The diagram below shows the shell mo of an atom.



Which of the following statements is correct?

I. The mass (nucleon) number is
II. The atomic (proton) number is
III. The atom shown in the diagra positively charged.

(A) I only
(B) II only
(C) I and II only
(D) I, II and III

58. When alpha particles, beta particles gamma rays from a radioactive sourc subjected to a transverse magnetic which of them would NOT be de from their original path?

I. Alpha
II. Beta
III. Gamma

(A) II only
(B) III only
(C) I and II only
(D) II and III only

59. Which of the following statements about the expression $E = mc^2$ would be true?

I. It defines the maximum energy that light can have.
II. It represents the fact that mass can be converted into energy.
III. It represents one of the major achievements of Rutherford.

(A) II only
(B) I and II only
(C) II and III only
(D) I, II and III

60. What is the mass number, A, and atomic number, Z, of the new element formed when an alpha particle is emitted from radium $^{226}_{88}Ra$?

	A	Z
(A)	222	86
(B)	224	84
(C)	228	92
(D)	230	90

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.